

Reliability and Radiation Response of Magnetic Tunnel Junction Technologies

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The radiation response of magnetic tunnel junctions (MTJs) was characterized for spin-transfer torque MRAM (STT-MRAM), spin-orbit torque MRAM (SOT-MRAM), and voltage-controlled magnetic anisotropy MRAM (VCMA-MRAM). All MTJs showed decreases in coercivity and tunneling magnetoresistance (TMR) at high levels of displacement damage. VCMA devices show variation in coercivity at elevated temperatures after irradiation for some biases, but no difference in TMR. MTJs exhibit a high radiation tolerance and are a promising choice for resilient, high performance applications.

Index Terms—Displacement damage, total ionizing dose, magnetic tunnel junction, heavy ion irradiation

I. INTRODUCTION

THE DEMAND for non-volatile memory (NVM) has been rapidly increasing, driven by growth in fields like artificial intelligence and the internet of things, and there have been multiple emerging technologies that promise higher performance in metrics like speed, endurance, and energy efficiency, although Flash memory still has the highest density. The demand for NVM that can operate in radiation environments has increased as well due to factors like the growth in the commercial space sector. Magnetic random-access memory (MRAM) is an emerging type of NVM that has also shown promising performance metrics, radiation tolerance, and better scaling potential than other emerging NVM candidates.

Magnetic tunnel junctions (MTJs) allow the modulation of tunneling resistance based on whether the orientation of the magnetization of two ferromagnetic layers are aligned in parallel or anti-parallel states relative to each other [1]. This property is referred to as tunneling magnetoresistance (TMR), which is defined as the ratio of the difference in resistance values to the minimum resistance, and makes MTJs a key element of magnetic NVM technologies. MTJs typically have one ferromagnetic layer that has its magnetization fixed and one ferromagnetic layer that can have its magnetization orientation changed by applied voltages and currents, generally referred to as the free layer. This work will review investigations performed on spin-transfer torque MRAM (STT-MRAM), spin-orbit torque MRAM (SOT-MRAM), and voltage-controlled magnetic anisotropy MRAM (VCMA MRAM).

NVMs that operate in radiation environments are subjected to a variety of damage mechanisms. While it is necessary to evaluate radiation effects on both the NVM and the peripheral circuitry, this work focuses on the radiation response of MTJs, the fundamental building blocks of the memory. Two of the fundamental damage mechanisms are total ionizing dose (TID), where electron-hole pairs are generated inside semiconductor devices, and displacement damage (DD), where atoms are displaced from their normal positions in a lattice. MTJ based

NVMs store their state as a function of magnetic domain, which intrinsically makes them less susceptible to the alteration of their state from electron-hole pair generation than previous NVMs that stored their state as a function of charge. This work will review investigations into the radiation response of STT-MRAM cells and cells for several emerging types of MRAM.

II. EXPERIMENTAL DETAILS

Several radiation sources were used to study TID and DD on the various MTJs. DD was studied at Sandia's Ion Beam Laboratory using beams that were customized for each device to maximize displacements in the MTJ using stopping range of ions in matter (SRIM) calculations [SRIM ref]. A Co-60 Shepherd cell was used to study TID effects. Device details can be found in [2-4].

III. RESULTS

A. STT-MRAM

The STT-MRAM write mechanism uses spin polarized current running through the MTJ, which decreased the write current, contributing to better scalability. MTJs with perpendicular magnetic anisotropy (PMA) and STT switching were irradiated with 3 MeV Ta ions up to fluences ranging from 10^9 to 10^{14} cm⁻² [2]. Gradual decreases in device parameters were observed above fluences of 10^{11} cm⁻². More than ten devices were irradiated for each fluence and the observed trends were systematic. Fig. 1 plots the hysteresis loops, showing how the switching characteristics degrade with fluence. The maximum resistance and the coercive field begin to degrade at 10^{12} cm⁻², with the loop collapsing at a fluence of 10^{13} cm⁻² with a complete loss of TMR. Other studies have shown that TID has little impact on the state of MTJs in STT-MRAM [5].

B. SOT-MRAM

SOT-MRAM changes state via spin-orbit interactions generated by a spin polarized current through a heavy metal. MTJs with PMA and SOT switching were irradiated with 800 keV Ta ions to study DD and irradiated up to 1 Mrad(Si) with a Co-60 source to study TID [3]. No effects from TID were

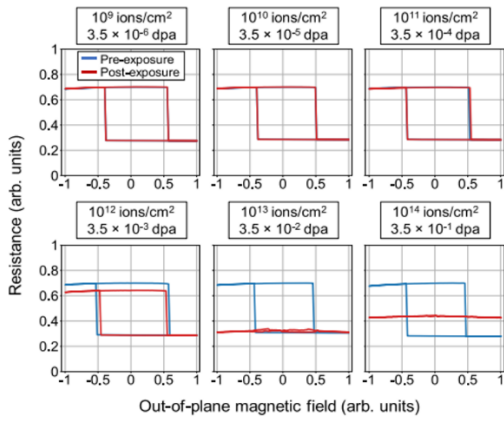


Fig. 1. Hysteresis loops of MTJ resistance pre- and post-irradiation with 3 MeV Ta ions [2].

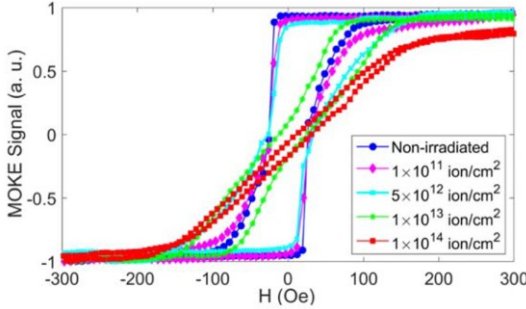


Fig. 2. Out-of-plane magnetic hysteresis loops characterized with magneto-optical Kerr effect (MOKE) for pre- and post-irradiation with 800 keV Ta ions [3].

observed after the Co-60 irradiation. Similarly to STT-MRAM, minimal changes were observed due to DD up to fluences of 10^{12} cm^{-2} , but at higher fluences coercivity and PMA degrade, as shown in Fig. 2.

C. VCMA-MRAM

Voltage pulses can alter the PMA of VCMA devices, enabling switching via electric field. VCMA MTJs with a range of different applied biases were irradiated with 20 MeV Au ions up to a fluence of $2 \times 10^{13} \text{ cm}^{-2}$ and coercivity and TMR both degraded above a fluence of 10^{12} cm^{-2} , as seen in Fig. 3 [4]. The degradation depended on the bias applied, with lower biases showing less degradation. Elevated temperatures may affect VCMA performance [6], so TMR and coercivity were characterized as a function of temperature pre- and post-irradiation. TMR showed no significant changes, but coercivity was negatively impacted for some biases.

D. MRAM Displacement Damage Comparison

The fluences used for each of the ion beams can be converted to a metric of damage called displacements per atom (DPA), which allows the various experiments to be compared on the same scale, regardless of the ion type and energy used for the irradiation. Fig. 4 plots the TMR degradation versus DPA for STT, SOT, and VCMA devices. The data for each type show good agreement, indicating that there is no significant difference in DD degradation CoFeB/MgO/CoFeB MTJs optimized for different switching mechanisms. Future work will address how write mechanisms are affected.

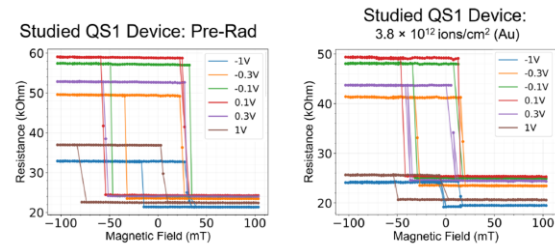


Fig. 3. Hysteresis loops at different bias levels for pre- and post-irradiation with 20 MeV Ta ions [4].

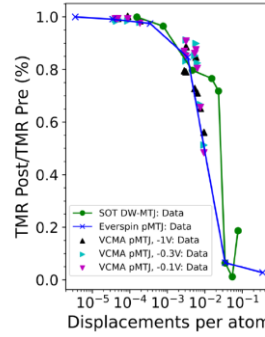


Fig. 4. Comparison of degradation due to DD across multiple MTJ devices [4]

IV. DISCUSSION AND CONCLUSION

Multiple types of MTJs have shown resilience to radiation damage, demonstrating intrinsic radiation-hardness. While all the MTJs studied thus far eventually degrade due to DD, the degradation occurs at extremely high levels. TID studies have shown little effect on MTJ characteristics either. Additional studies are needed to characterize any potential impacts to reliability metrics like endurance and retention due to radiation. However, the radiation results are promising and MTJ based NVM technology will likely be a strong candidate for applications in radiation environments.

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